

PRATHEEN REDDY PININITI

AI / ML Engineer | Generative AI • Agentic AI • Computer Vision • Applied ML

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PROFESSIONAL SUMMARY

AI/ML Engineer and M.S. Artificial Intelligence candidate (GPA 3.9, Florida Atlantic University, May 2026) with nearly 2 years of startup engineering experience (frontend, GitHub Actions CI/CD, product growth analytics) and hands-on work building agentic AI systems, generative AI products, and computer vision pipelines — owning delivery from Python data pipelines and CI/CD through React and React Native interfaces.

TECHNICAL SKILLS

Generative AI & LLMs: LLaMA, Mistral, Gemma, Prompt Engineering, RAG, Vector Databases (pgvector), LoRA/QLoRA Fine-Tuning, LLM Evaluation & Guardrails
Agentic AI: LangChain, LangGraph, ReAct Agents, Tool Use, Multi-Agent Orchestration, State Management
Computer Vision: CLIP, SAM-2, MediaPipe, ArcFace, IP-Adapter FaceID, Real-ESRGAN, NAFNet, YOLOv8, Virtual Try-On (FASHN VTON)
Machine Learning & Data: Python, PyTorch, TensorFlow, scikit-learn, XGBoost, PCA/ICA, Signal & Noise Analysis, NumPy, Pandas, SciPy, SQL
Deployment & Cloud: CI/CD (GitHub Actions), Docker, Kubernetes (basic), Git, Google Cloud (GCP), Azure, AWS (Bedrock, SageMaker), FastAPI, Supabase, Firebase, BigQuery
Interfaces: React, React Native, TypeScript, Three.js, Flutter

PROFESSIONAL EXPERIENCE

Founder & AI Engineer — Auraltr (Fashion Tech Startup) May 2026 – Present

Self-Directed Startup — Miami, FL

- Built a GenAI virtual try-on MVP in 8 weeks (React Native, CLIP embeddings, Supabase), combining computer vision with LLM-based outfit recommendations.
- Raised identity preservation to 92% across 25 test users by building a face-identity pipeline with MediaPipe, ArcFace, and IP-Adapter FaceID.
- Improved garment retrieval relevance to 88% top-5 accuracy by indexing CLIP embeddings in pgvector for semantic search and affiliate matching.
- Moved try-on inference from a paid API to the open-source FASHN VTON v1.5 model, removing per-try-on API fees.

Graduate Research Assistant — AI & Imaging Systems Jan 2025 – May 2026

Florida Atlantic University, in collaboration with Canon Inc. — Boca Raton, FL

- Cut sensor characterization time by roughly 60% versus manual methods by building a 5-tier Python ML pipeline (dark noise, PRNU, inter-pixel correlation, PSD, PCA/ICA) for the Canon MS-500 SPAD sensor.
- Automated iterative sensor diagnostics with a LangChain + ReAct agentic proof of concept, removing roughly 6 hours of manual analysis per week.
- Isolated 5 distinct noise sources in the DN domain, giving Canon stakeholders a quantified characterization baseline published as a formal thesis.
- Standardized analysis across 4 researchers with reusable playbooks, aligned with Canon engineers under Responsible AI principles.

Teaching Assistant — Machine Learning & Communication Networks Jan 2025 – May 2026

Florida Atlantic University — Boca Raton, FL

- Facilitated machine learning labs (regression, classification, deep learning) for 60+ students and built reusable curriculum assets adopted across 3 course sections.

Software Engineer — TinyPal Oct 2022 – Jun 2024

TinyPal Parenting Pvt. Ltd. (AI parenting-app startup; joined during pre-launch R&D) — Hyderabad, India (Hybrid)

- Developed frontend features in React and TypeScript for an AI-powered parenting app built around expert-curated child-development goals, a personalized feed, and an AI parenting assistant.
- Cut deployment latency by 25% by building and restructuring CI/CD workflows in GitHub Actions to automate builds and deployments.
- Worked across Google Cloud (GCP) and Azure, connecting the app's services between the two clouds with a lightweight Kubernetes setup.
- Collected user and product data and analyzed the app's growth and engagement trends, sharing findings with the team to guide feature priorities.

PROJECTS

Warroom — Multi-Agent AI Platform | LangGraph, ReAct, RAG 2025

- Completed 80% of benchmark business-automation tasks without human intervention by orchestrating ReAct agents with LangGraph state management and RAG retrieval.
- Reduced new-workflow setup from about 4 hours to under 1 hour with reusable agent playbooks and a shared tool schema.

NeuroPulse — Stroke Risk Prediction App | XGBoost, Three.js, Blender 2025

- Reached 0.89 AUC and 82% recall on held-out data with an XGBoost clinical risk model, with an explainability layer surfacing the top 5 features per prediction behind a 3D brain visualization.

Signal (IdeaCloud) — On-Device AI Voice Capture | Kotlin, Jetpack Compose, Gemma 2026

- Auto-generated titles, summaries, and categories for 85% of captured ideas using a local Gemma model, keeping 100% of audio on-device.

EDUCATION

M.S. Artificial Intelligence — Florida Atlantic University, Boca Raton, FL | GPA 3.9 Aug 2024 – May 2026

- Thesis: SPAD Camera Image Analysis — DN-Domain Noise Characterization of the Canon MS-500 SPAD Sensor

B.Tech Computer Science (AI & ML) — Lovely Professional University, India | GPA 3.7 May 2024

CERTIFICATIONS & PUBLICATIONS

- Thesis Published: SPAD Camera Image Analysis — Canon MS-500 (2026)
- Google Advanced Data Analytics Professional Certificate (2025)
- Google UX Design Professional Certificate (2025)